

Heat and Mass Transfer

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Syllabus

CE3347 - B.Tech II Degree Course – Scheme of Examinations – A Syllabus 2006 ME Sem IV

ME 604 HEAT AND MASS TRANSFER

Module I

Introduction to heat transfer – basic modes of heat transfer – conduction heat transfer – energy balance – integral and differential approaches – general heat conduction equation in Cartesian, cylindrical and spherical coordinates – initial and boundary conditions – one-dimensional steady state conduction with heat generation – conduction shape factor – temperature dependence of thermal conductivity – applications like extended surface heat transfer and critical insulation thickness – two dimensional steady state heat conduction – examples – unsteady state heat conduction in one dimension – lumped heat capacity system – semi-infinite solids with sudden and periodic change in surface temperature – numerical methods in conduction problems.

Module II

Convective heat transfer – Newton's law of cooling – thermal boundary layer – Prandtl number – hydrodynamic and thermal boundary layer equations – laminar forced convection heat transfer from flat plates – internal flow and heat transfer – fully developed laminar flow in pipes – turbulent forced convection – Reynolds' analogy – natural convection – natural convection heat transfer from vertical plates – condensation and boiling – film and drop wise condensation – film boiling and pool boiling – introduction to multiphase flow and heat transfer. Diffusion and convective mass transfer.

Module III

Radiative transfer – electromagnetic radiation spectrum – thermal radiation – black body, grey body and coloured body – monochromatic and total emissive power – Planck's law – Stefan-Boltzman law – Wien's displacement law – absorptivity – reflectivity – transmissivity – emissivity – Kirchhoff's identity – radiation exchange between surfaces – shape factor – shape factors for simple configurations – heat transfer in the presence of re-radiating surfaces – radiation shields – surface and shape resistances – electrical network analogy.

Module IV

Heat Exchangers: Type of heat exchangers, Logarithmic mean temperature difference (LMTD), analysis using fouling factors, derivation of LMTD for parallel flow and counter flow heat exchangers, effectiveness, NTU method of heat exchanger analysis, fouling factors, simple design problems.

TEXT BOOK

1. Cengel Heat Transfer Tata McGraw Hill
2. Holman J.P., "Heat Transfer", McGraw Hill International Students Edition

REFERENCE BOOKS

1. Incorpera F.P. & De Witt D.P., "Fundamentals of Heat and Mass Transfer", John Wiley
2. Kreith F., "Heat Transfer", International Text Book Company
3. Gebhart B., "Heat Transfer", McGraw Hill
4. R.K. Rajput, "Heat and Mass Transfer"
5. Goshdostidar, "Heat and Mass Transfer"

Type of questions for University Examination

Question 1 - 8 short answer questions of 5 marks each, 2 questions from each module

Question 2-5 - There will be two choices from each module. Answer one question from each module of 15 marks

Introduction

- Heat exchanger - can be defined as any device that transfers heat from one fluid to another or from or to a fluid and the environment.
- Classification according to transfer process
 - (i) Direct contact type-Gas -liquid, Liquid vapor
 - (ii) Indirect contact type - Storage type, Fluidized bed
- Classification according to surface compactness
 - (i) Non -Compact $< 400m^2/m^3$
 - (ii) Compact $> 400m^2/m^3$
- Classification according to construction
 - (i) Tubular - Double pipe, shell and tube, pipe coils
 - (ii) Plate type - PHE, spiral
 - (iii) Extended surface - Plate fin, Tube fin
 - (iv) Regenerative - Rotory, Fixed matrix
- Classification according to flow arrangements
 - (i) Single pass- Counter flow, parallel flow, cross flow
 - (ii) Multi pass - shell and tube m-shell pass, n-tube pass, plate

Heat Exchangers



Applications

Process Industry - Refinery, Petrochemicals, Dairy, Food Industries,
Air Conditioning

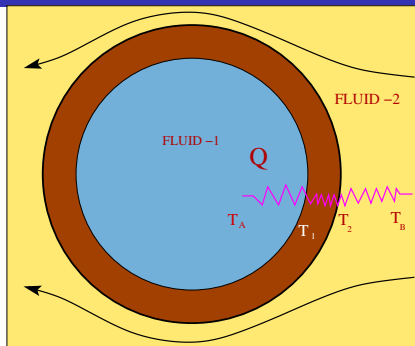
Heat Exchangers



Applications

Process Industry - Refinery, Petrochemicals, Automobiles, Power plants, Air conditioning, Boilers, Furnaces

Overall Heat Transfer Coefficient



Heat transfer through tube

- Heat transfer from Fluid A to Tube material

$$Q = h_A A_i (T_A - T_1),$$
- h_A - Tube inside convection heat transfer coefficient, Tube inside surface area $A_i = \pi D_i L$

- Heat transfer through the tube material $Q = \frac{(T_1 - T_2)2\pi kL}{\ln\left(\frac{r_o}{r_i}\right)}$
- Heat transfer from Tube to Fluid B, $Q = h_B A_o (T_2 - T_B)$,
- h_B - Tube outside convection heat transfer coefficient, Tube outside surface area $A_o = \pi D_o L$
- From the above three eqns $Q = \frac{T_A - T_B}{\left(\frac{1}{h_i A_i} + \frac{\ln\left[\frac{r_o}{r_i}\right]}{2\pi kL} + \frac{1}{h_o A_o}\right)}$
- Heat transfer $Q = \frac{A_i (T_A - T_B)}{\left(\frac{1}{h_i} + \frac{A_i \ln\left[\frac{r_o}{r_i}\right]}{2\pi kL} + \frac{A_i}{h_o A_o}\right)}$
- We define U_i The overall heat transfer coefficient based on inside surface area as $U_i = \frac{1}{\left(\frac{1}{h_i} + \frac{A_i \ln\left[\frac{r_o}{r_i}\right]}{2\pi kL} + \frac{A_i}{h_o A_o}\right)}$
- The heat transfer $Q = U_i A_i (T_A - T_B)$

- Heat transfer $Q = \frac{A_o(T_A - T_B)}{\left(\frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi k L} + \frac{1}{h_o} \right)}$
- We define U_o The overall heat transfer coefficient based on outside surface area as $U_o = \frac{1}{\left(\frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi k L} + \frac{1}{h_o} \right)}$
- The heat transfer $Q = U_o A_o (T_A - T_B)$

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- The heat transfer $Q = U_o A_o (T_A - T_B)$
- U_o and U_i different ?, what happens to A_i and A_o ?

Fouling in Heat exchangers



Fouling

- Fouling - due to deposition of dirt, mud, contaminants, corrosion, erosion, etc.
- Fouling on both sides -inside the tube and outside the tubes.
- Deteriorate heat exchanger performance.

Fouling considerations - Design

- Fouling factor (K/W)
- Fouling factor to be considered during design ?
- High or Low value for Fouling factor ?
- Area requirement with fouling considerations
- Overall heat transfer coefficient with Fouling factor
- How to get fouling factor values ? [► Foul Fact](#)
- Velocity considerations

Fouling in Heat exchangers

Typical Fouling Factors [$\text{m}^2\text{K/W}$]

Cooling Water

Conditions

cooling water < 50 C
cooled fluid < 120 C

cooling water > 50 C
cooled fluid > 120 C

Cooling Water velocity

$v < 1 \text{ m/s}$

$v > 1 \text{ m/s}$

$v < 1 \text{ m/s}$

$v > 1 \text{ m/s}$

Type of Water

Sea	0.00009	0.00009	0.00018	0.00018
Brackish	0.00035	0.00018	0.00053	0.00035
Cooling tower with inhibitor	0.00018	0.00018	0.00035	0.00035
Cooling tower without inhibitor	0.00053	0.00053	0.00088	0.0007
City grid	0.00018	0.00018	0.00035	0.00035
River minimum	0.00018	0.00018	0.00035	0.00035
River average	0.00053	0.00035	0.0007	0.00035
Engine jacket	0.00018	0.00018	0.00018	0.00018

Fouling

- Type of fluid, Fluid velocity, Temperature
- Turbulators, Service conditions

Overall Heat transfer coefficient U

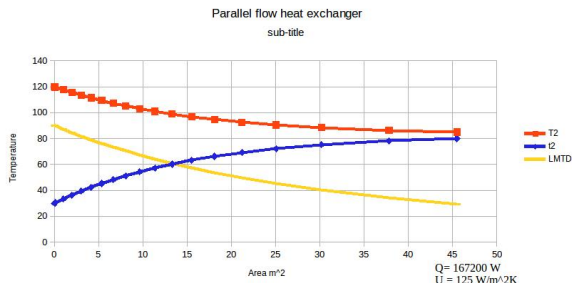
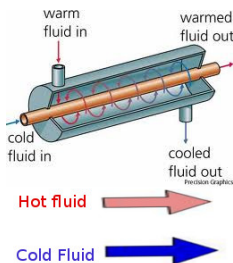
- Overall heat transfer coefficient

$$U_i = \frac{1}{\left(F_i + \frac{1}{h_i} + \frac{A_i \ln \left[\frac{r_o}{r_i} \right]}{2\pi kL} + \frac{A_i}{h_o A_o} + \frac{F_o A_i}{A_o} \right)}$$

- Overall heat transfer coefficient

$$U_o = \frac{1}{\left(\frac{A_o F_i}{A_i} + \frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi kL} + \frac{1}{h_o} + F_o \right)}$$

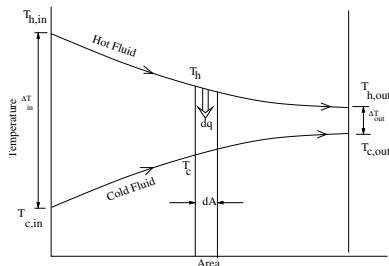
Parallel flow heat exchangers



Parallel flow heat exchangers

- Both fluid flow in same direction
- Temperature difference between fluids reduce towards exit
- Heat transfer $Q = U_i A_i \Delta T$ or $Q = U_o A_o \Delta T$ - Which ΔT ?

Heat Transfer in Parallel Flow Heat exchangers



- Hot fluid entering temperature at $T_{h,in}$, Exit temperature $T_{h,out}$
- Cold fluid entering temperature at $T_{c,in}$, Exit temperature $T_{c,out}$
- Heat duty of the Heat exchanger $Q = U_o A_o \Delta T = U_i A_i \Delta T$
- Also $Q = m_h c_{ph} (T_{h,in} - T_{h,out}) = m_c c_{pc} (T_{c,out} - T_{c,in})$

Log Mean Temperature Difference LMTD

- Heat transfer within the elemental area dA
 $-dq = U(T_h - T_c)dA$
- for the heat transfer $dq = -m_h c_h dT_h = m_c c_c dT_c$
- From the heat transfer eqn $dT_h = \frac{-dq}{m_h c_h}$ and $dT_c = \frac{dq}{m_c c_c}$
- Thus, $dT_h - dT_c = d(T_h - T_c) = -dq \left(\frac{1}{m_h c_h} + \frac{1}{m_c c_c} \right)$
- Substituting for dq gives $\frac{d(T_h - T_c)}{T_h - T_c} = -U \left(\frac{1}{m_h c_h} + \frac{1}{m_c c_c} \right) dA$
- Integrating we get,

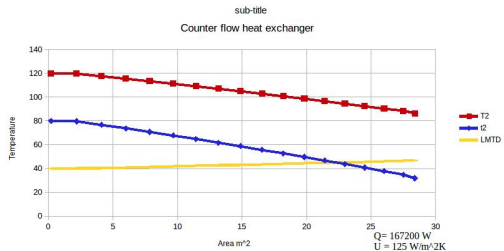
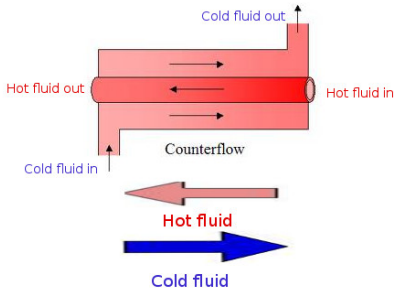
$$\ln \left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = -UA \left(\frac{1}{m_h c_h} + \frac{1}{m_c c_c} \right) \quad (1)$$

Log Mean Temperature Difference LMTD

- We have, $m_h c_h = \frac{Q}{T_{h,in} - T_{h,out}}$ and, $m_c c_c = \frac{Q}{T_{c,out} - T_{c,in}}$
- Substituting these in Eq (1)

$$\ln \left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = -UA \left(\frac{[T_{h,in} - T_{h,out}]}{Q} + \frac{[T_{c,out} - T_{c,in}]}{Q} \right)$$
- Re-arranging $Q = UA \frac{[T_{h,out} - T_{c,out}] - [T_{h,in} - T_{c,in}]}{\ln \left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right)}$
- LMTD is defined as $\Delta T_m = \frac{[T_{h,out} - T_{c,out}] - [T_{h,in} - T_{c,in}]}{\ln \left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right)}$
- LMTD, $\Delta T_m = \frac{\Delta T_{in} - \Delta T_{out}}{\ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right)}$

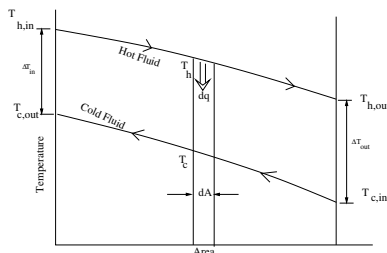
Counter flow heat exchangers



Counter flow heat exchangers

- Fluid flow in opposite direction
- Temperature difference of the fluids reduce
- Heat transfer $Q = U_i A_i \Delta T$ or $Q = U_o A_o \Delta T$ - Which ΔT ?

Heat Transfer in Counter Flow Heat exchangers



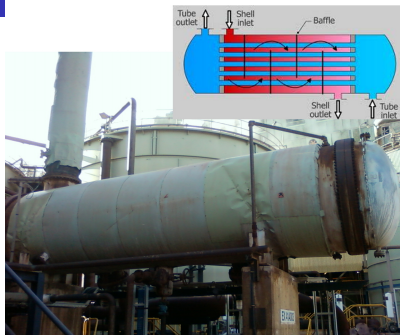
- Hot fluid entering temperature at $T_{h,in}$, Exit temperature $T_{h,out}$
- Cold fluid entering temperature at $T_{c,in}$, Exit temperature $T_{c,out}$
- Heat duty of the Heat exchanger $Q = U_o A_o \Delta T = U_i A_i \Delta T$
- Also $Q = m_h c_{ph} (T_{h,in} - T_{h,out}) = m_c c_{pc} (T_{c,out} - T_{c,in})$

- LMTD for counter flow heat exchanger

$$\Delta T_m = \frac{[T_{h,in} - T_{c,out}] - [T_{h,out} - T_{c,in}]}{\ln \left(\frac{T_{h,in} - T_{c,out}}{T_{h,out} - T_{c,in}} \right)} \quad (2)$$

- LMTD, $\Delta T_m = \frac{\Delta T_{in} - \Delta T_{out}}{\ln \left(\frac{\Delta T_{in}}{\Delta T_{out}} \right)}$
- In a case where $\Delta T_{in} = \Delta T_{out}$, The LMTD, $\Delta T_m = \Delta T_{in} = \Delta T_{out}$

Shell and Tube Heat Exchangers



Tubes

Available in Copper, 90/10 Cu/Ni, 316 Stainless Steel, Admiralty or Carbon Steel. Tubes are roller expanded.

Shells

Rugged shell available in Steel and 316 Stainless Steel. Minimum clearances between shell and baffles reduce by-pass and maximize heat transfer.

Heads

Heads available in Cast Iron, Brass, 316 Stainless Steel or Fabricated Carbon Steel.

Tube sheets

Thick Carbon Steel, 316 Stainless Steel or 90/10 Cu/Ni tube sheets.

Supports

Movable, fabricated Carbon Steel available on all sizes.

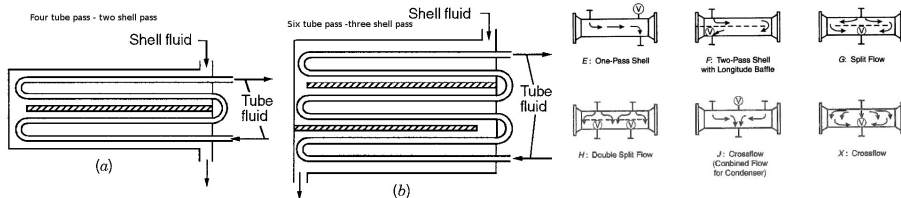
Baffles

Precision punched baffles assure effective circulation by providing minimum clearances between the tubes and tube holes. Baffle cuts and spacing for each diameter are consistent with best practice. Standard available material includes Carbon Steel, Brass and 316 Stainless Steel.

Shell and tube

- One fluid on the tube side and other fluid on the shell side
- Widely used type of heat exchangers in the industry
- Easy to clean, repair and maintenance
- International design codes TEMA, ASME

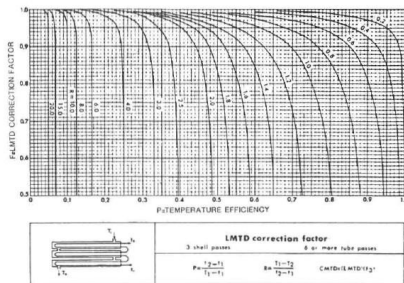
Shell and Tube Heat Exchangers Design



Design

- Assume the heat exchanger is counter flow heat exchanger
- Find the LMTD ΔT_m using Eq. (19)
- Find the correction factor from graph F corresponding to the type of shell and tube heat exchanger
- $Q = UAF\Delta T_m$

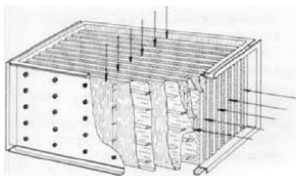
Shell and Tube Heat Exchangers Design



LMTD Correction

- Find the number of tube pass, shell pass
- Choose the graph
- Find $P = \frac{T_{c,out} - T_{c,in}}{T_{h,in} - T_{c,in}}$ and $R = \frac{T_{h,in} - T_{h,out}}{T_{c,out} - T_{c,in}}$
- Find the value F and Heat transfer $Q = UAF\Delta T_m$
- $F = 1$ for boiling or condensation

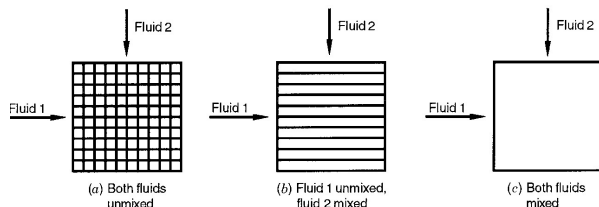
Cross Flow Heat Exchangers



Cross

- One fluid on one side and other fluid on the other side flow in a cross pattern
- Widely used type of heat exchangers in the industry, space, automobile, Air conditioners, etc.
- face difficulty to clean, repair and maintenance
- Mostly compact type of heat exchangers

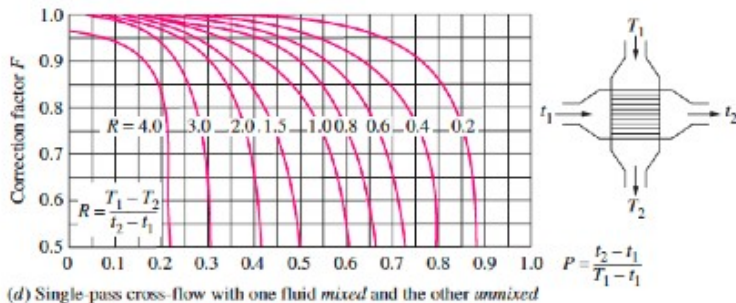
Cross Flow Heat Exchangers Design



Design

- Assume the heat exchanger is counter flow heat exchanger
- Find the LMTD ΔT_m using Eq. (19)
- Find the correction factor from graph F corresponding to the type of cross flow heat exchanger
- $Q = UAF\Delta T_m$, $F = 1$ for boiling or condensation

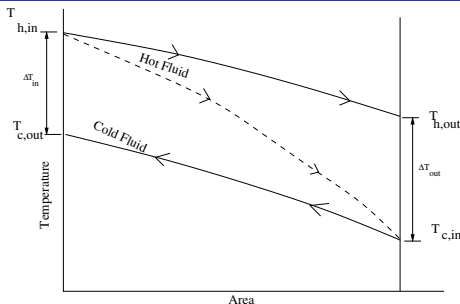
Cross Flow Heat Exchangers Design



LMTD Correction

- Find the type - Fluid mixed or un-mixed
- Choose the graph
- Find $P = \frac{T_{c,out} - T_{c,in}}{T_{h,in} - T_{c,in}}$ and $R = \frac{T_{h,in} - T_{h,out}}{T_{c,out} - T_{c,in}}$
- Find the value F and Heat transfer $Q = UAF\Delta T_m$

Effectiveness -NTU Method



- Heat Exchanger Sizing and Rating
- All Four Temperature of fluid may not be available
- Maximum temp. diff. possible $T_{h,out}$ can $\Rightarrow T_{c,in}$
- Heat capacity of fluids $m_h c_h = C_h$ and $m_c c_c = C_c$

- One heat capacity is minimum C_{min} and other maximum C_{max}
- Maximum heat transfer $C_{min} \times$ maximum temp diff.
- Maximum heat transfer $= C_{min}(T_{h,in} - T_{c,in})$
- Why C_{min} not C_{max} ?

- One heat capacity is minimum C_{min} and other maximum C_{max}
- Maximum heat transfer $C_{min} \times$ maximum temp diff.
- Maximum heat transfer $= C_{min}(T_{h,in} - T_{c,in})$
- Why C_{min} not C_{max} ?
- Will cause temperature crossing
- Effectiveness $\epsilon = \frac{\text{Actual Heat Transfer}}{\text{Maximum possible Heat Transfer}}$
- From Eq (1), We have $\ln \left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = -UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)$
- $\left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$
- We have $C_h(T_{h,in} - T_{h,out}) = C_c(T_{c,out} - T_{c,in})$
- $T_{h,out} = T_{h,in} - \frac{C_c}{C_h}(T_{c,out} - T_{c,in})$
- Substituting $\left(\frac{T_{h,in} - \frac{C_c}{C_h}(T_{c,out} - T_{c,in}) - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

- Re-arranging $\left(\frac{T_{h,in} - \frac{C_c}{C_h} T_{c,out} + \frac{C_c}{C_h} T_{c,in} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

- $\left(\frac{T_{h,in} - T_{c,in} + T_{c,in} - \frac{C_c}{C_h} T_{c,out} + \frac{C_c}{C_h} T_{c,in} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

- $\left(\frac{T_{h,in} - T_{c,in} + T_{c,in} \left(1 + \frac{C_c}{C_h} \right) - \left(1 + \frac{C_c}{C_h} \right) T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

-

$$1 - \left[\frac{T_{c,out} - T_{c,in}}{T_{h,in} - T_{c,in}} \right] \left(1 + \frac{C_c}{C_h} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)} \quad (3)$$

- We have Effectiveness $\epsilon = \frac{\text{Actual Heat Transfer}}{\text{Maximum possible Heat Transfer}}$

- Actual heat transfer

$$Q_{ac} = C_h(T_{h,in} - T_{h,out}) = C_c(T_{c,out} - T_{c,in})$$

- Maximum heat transfer $Q_{max} = C_{min}(T_{h,in} - T_{c,in})$

- Effectiveness $\epsilon = \frac{C_c(T_{c,out} - T_{c,in})}{C_{min}(T_{h,in} - T_{c,in})}$
- Re-arranging $(T_{c,out} - T_{c,in}) = \epsilon \frac{C_{min}}{C_c} (T_{h,in} - T_{c,in})$
- Substituting in Eq (3)
- $1 - \epsilon \frac{C_{min}}{C_c} \left(1 + \frac{C_c}{C_h}\right) = e^{-UA\left(\frac{1}{C_h} + \frac{1}{C_c}\right)}$
- $\epsilon = \frac{1 - e^{-UA\left(\frac{1}{C_h} + \frac{1}{C_c}\right)}}{\frac{C_{min}}{C_c} \left(1 + \frac{C_c}{C_h}\right)}$
- $\epsilon = \frac{1 - e^{-UA\left(\frac{1}{C_h} + \frac{1}{C_c}\right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h}\right)}$

$$\epsilon = \frac{1 - e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h} \right)}$$

$$\epsilon = \frac{1 - e^{-\frac{UA}{C_{min}} \left(\frac{C_{min}}{C_h} + \frac{C_{min}}{C_c} \right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h} \right)}$$

■ We define Number of heat transfer units (NTU) $N = \frac{UA}{C_{min}}$

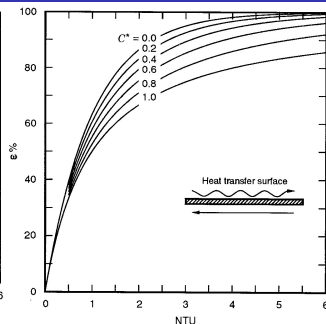
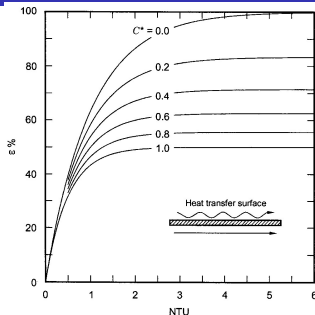
$$\epsilon = \frac{1 - e^{-N \left(\frac{C_{min}}{C_h} + \frac{C_{min}}{C_c} \right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h} \right)}$$

■ We define $C = \frac{C_{min}}{C_{max}}$

■ Out of C_h and C_c One will be maximum and the other will be minimum

■ Thus for parallel flow, $\epsilon = \frac{1 - \exp[-N(1+C)]}{1+C}$

Effectiveness -NTU Method Rating problem



- Find NTU from U , Area and C_{min}
- From the graph corresponding to the type of HE find ϵ
- Using ϵ we can find the exit temperature

Heat Exchanger Design -Remarks

- Significance of NTU
- Difference between LMTD method and Effectiveness-NTU method
- Heat exchanger design considerations
- Pressure drop, materials
- Type of service
- Heat Exchanger optimization
- Heat Exchanger Design Softwares